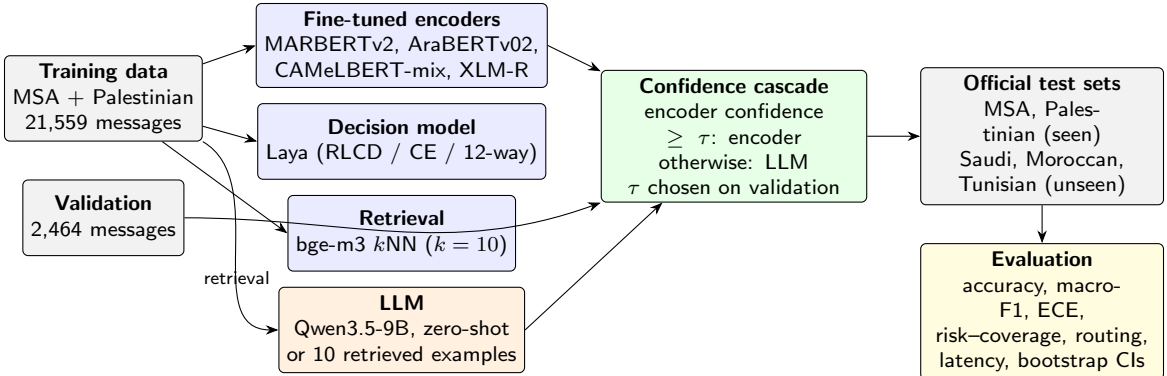
Training data
MSA + Palestinian
21,559 messages
Fine-tuned encoders
MARBERTv2, AraBERTv02,
CAMeLBERT-mix, XLM-R
Decision model
Laya (RLCD / CE / 12-way)
Validation
2,464 messages
Retrieval
bge-m3 kNN ($k = 10$)
retrieval
LLM
Qwen3.5-9B, zero-shot
or 10 retrieved examples
Confidence cascade
encoder confidence
$\geq \tau$: encoder
otherwise: LLM
τ chosen on validation
Official test sets
MSA, Palestinian (seen)
Saudi, Moroccan,
Tunisian (unseen)
Evaluation
accuracy, macro-F1, ECE,
risk–coverage, routing,
latency, bootstrap CIs